INCH-POUND

MIL-H-14253F(AL)
10 June 1994
SUPERSEDING
MIL-H-14253E(AL)
20 May 1985

MILITARY SPECIFICATION

HONING MACHINE, HORIZONTAL, WET TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a horizontal honing machine, bench mounted, for the honing of bushings, bearings, and hydraulic brake cylinders including step cylinders. The honing machine is to be used primarily as a wet type.However, the honing machine may also be used as a dry type, when required.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement therto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898	- Cushioning Material, Cellulosic
L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polypolefin
W-C-596/13	- Connector, Plug, Electrical, General Purpose, Hospital Grade Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3419

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PPP-B-601 – Boxes, Wood, Cleated-Plywood
PPP-B-621 – Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 – Boxes, Shipping, Fiberboard
PPP-B-640 – Boxes, Fiberboard, Corrugated, Triple Wall
PPP-D-705 – Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity
PPP-D-729 – Drums, Shipping and Storage, Steel 55 Gallon (208 Liters)
PPP-P-40 – Preservation and Packing of Hand Tools; Tool and Tool Accessories for Power Driven, Metal and Woodworking Machinery
PPP-B-1055 – Barrier Material, Waterproof, Flexible
PPP-T-60 – Tape: Packaging, Waterproof

MILITARY

MIL-P-116 – Preservation, Methods of
MIL-B-121 – Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-H-775 – Hose, Hose Assemblies; Rubber, Plastic, Fabric, Or Metal (Including Tubing) and Fittings, Nozzles and and Strainers, Packaging of
MIL-C-5501 – Caps and Plugs, Protective, Dust and Moisture Seal
MIL-E-16298 – **Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of**
MIL-E-17555 – **Electronic and Electric Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of**
MIL-B-26195 – **Boxes, Wood-Cleated, Load-Bearing Base**

STANDARDS

FEDERAL

FED-STD-H28 – Screw–Threads Standards for Federal Services
FED-STD-101 – Test Procedures for Packaging Materials

MILITARY

MIL-STD-129 – Marking for Shipment and Storage
MIL-STD-889 – Dissimilar Metals
MIL-STD-1186 – Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods
MIL-STD-1190 Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications are available from the Defense Printing Service Detachment Office, Building 4D NPM-DoDSSP, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation (see 6.2).

US DEPARTMENT OF LABOR - CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1900 to 1910 - General Industry, OSHA Safety and Health Standards.

(Unless otherwise indicated, copies of the regulation are available from the Superintendent of Documents, Government Printing Office, Washington DC 20402).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, The issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.9 - Machines, Safety Requirements for the Construction, Care, and Use of

(Applications for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3363 - Standard Test Method for Film Hardness by Pencil Test

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery

(Application for copies should be addressed to the National Fire Protection Association, Battery March Street, Quincy, MA 02269.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG-1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 268 - Metric Practices

(Applications for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th. Street, New York, NY 10017.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents are also available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First Article. When specified (see 6.2), the contractor shall furnish one or more honing machines for first article inspection. The first article may be either a preproduction model(s) or an initial production item(s) which conforms to the requirements specified herein. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements specified herein.

3.2 Design. The honing machine shall be new and of the manufacturer's latest design which shall meet the requirements specified herein. The honing machine shall be designed for bench mounting to hone bushings, bearings, and hydraulic cylinders, including step clinders. The design shall be such that all parts subject to wear, adjustment, and replacement shall be accessible.

3.2.1 Safety and health requirements. Covers, guards, or other safety devices shall be provided. The devices shall not interfere with operation of the honing machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms and assemblies furnished on the honing machine, whether or not specifically required herein shall comply with the requirements of 29 CFR Parts 1900 to 1910 that are applicable to the honing machine itself. In addition the honing machine shall comply with all requirements of ANSI B11.9 that are designated therein as the responsibility of the honing machine manufacturer. In the event of conflict between the requirements herein and OSHA, OSHA shall take precedence.

3.2.2 Lubricating systems. AU bearings (except sealed for life, permanently lubricated type), matching gears, and all other movinng parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. All oil holes, grease fittings, and filler caps shall be readily accessible. The supplier shall install or include all lubricants, fluids, greases and other compounds (see 5.7.1.9).
procedure.

3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the honing machine. All replacement, repair and spare parts shall provide the same or better form, fit, function, and performance as the parts provided initially installed in the honing machine. When industry or Govern standards are applicable, all replacement parts shall conform to the standards.

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3.24 Measurement system. All measurements, dimensions, sizes and capacities in this document are given in the US system. These measurements may be converted to the SI system through the use of conversion factors and methods specified in IEEE 268.

3.3 Material. Materials not specifically designated herin or in the contract shall be of a quality commensurate with best commercial practices, shall be suitable for the intended purpose in the design and fabrication of the honing machine, and shall meet all requirements specified herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or logevity of the individual components or the honing machine. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The manufacturer may use reclaimed materials for the fabrication of new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4 Construction. The honing machine shall be complete so that when installed the honing machine can be used for the operations specified herein.

3.4.1 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.4.2 Threads. All machined threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.4.3 Painting and protective finish. Painting and protective finishing shall be in accordance with the best practice of the honing machine industry. The protective finish shall be a paint coating of at least 2.0 mils (0.002 inch) (0.0508 mm) thickness. All surfaces to be painted shall, immediately prior to painting, be cleaned, free of all foreign matter, and dried. Paint adhesion shall withstand removal of fiberglass tape firmly applied over two intersecting cuts in the paint without paint peeling from the base metal. The intersecting cuts shall each be two inches long, and to the base metal. Where noprotective finish is utilized (such as machine bedways, guide surfaces, plated surfaces, etc.) the natural finish of the material or finish obtained from heat treatment is permissible, provided the surfaces are free of scale and have a protective coating to inhibit corrosion. The surface hardness of the paint shall be between theindustrial standards of H to HB pencil lead as determined in accordance with ASTM D3363.

3.4.4 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, etc. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, for improving consmetic apperance.

3.4.5 Welding, brazing, and soldering. All welding, brazing, and soldering shall be free of defects and of a quality which shall sustain all the requirements of the welded, brazed, or soldered parts. Defects shall include but are not limited to such things as bubbles, cracks, gouges, holes, insufficient material used etc. All flux or

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other materials used during these operations shall be thoroughly removed(cleaned), from the part(s) immediatedly upon completion of the operation. These operations, shall not be employed as repair measures for defective parts.

3.5 Performance and product characteristics.

3.5.1 Honing machine. The honing machine shall be capable of either wet or dry honing operations without adding or removing special attachments. The honing machine shall be capable of honing through holes of the diameter size range of no less than 0.480 to 3 inches (12.192 to 76.2 mm) inclusive and for blind holes and step cylinders of the diameter size range of no less than 1.125 to 2.062 inches (28.575 to 52.375 mm) inclusive. The honing machine shall be capable of honing internal cylindrical surfaces to an accuracy of + 0.0002 inch (0.00508 mm) for roundness and taper within a diameter size range of no less than 0.480 to 3.000 inches (12.192 to 76.2 mm) inclusive. The hone shall produce a surface finish on gray iron to a roughness value no greater than 16 microinches (0.40 micrometers), arithmetical average (aa).

3.5.2 Spindle speed. The spindle speed shall be variable to perform the honing function within the required ranges and accuracy tolerance specified in 3.5.1.

3.5.3 Feed controls. The honing machine shall be equipped with a feed control mechanism. The mechanism shall control the cutting diameter of the honing mandrel stones. A means to regulate the cutting pressure of the mandrel stones shall also be provided.

3.5.4 Load controls. The honing machine shall be equipped with controls for stopping the mandrel by releasing the load without stopping the drive motor. If th honing machine is belt driven, a control device for belt tension and a belt guard shall be provided. A means for clutching action shall be provided so that when the mandrel is stopped, the load on the motor shall be completely released. The regulation of the cutting pressure of the mandrel stones and the clutching action shall be actuated by the same control.

3.6 Components.

3.6.1 Coolant system. A coolant system shall be provided that consists of a pump, a splash guard or similar device, a 3-gallon (11.355 liter) minimum capacity reservoir, and a baffling sustem for removing the honing wastes before reusing the coolant. The settled honing wastes shall be accessible for cleaning purposes. When a single drive motor is used for the spindle and pump drive, provisions shall be included to prevent the flow of coolant so that the honing machine can be used for dry honing.

3.6.2 Frame. All parts of the honing machine shall be mounted on a common frame. Provision shall be made for bench mounting the honing machine with not less than two bolts.

3.6.3 Electrical equipment. Unless otherwise specified (see 6.2), all electrical components, including the starters, relays, switches, and wiring, shall conform to, and shall be located in accordance with NFPA 79.

3.6.3.1 Motors. An electric rotor shall be provided that is a continuous duty induction squirrel-cage type conforming to the requirements of NEMA MG-1 with grade B insulation or better. Unless otherwise specified (see 6.2), the rotor shall operate on 110/220 volt, 50/60 hertz, single phase current. The spindle drive rotor shall no less than 1/3 horsepower (hp).

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3.6.3.2 Switch. An OFF/ON switch shall be provided for starting and stopping each motor.

3.6.3.3 Power supply cord. Unless otherwise specified (see 6.2), the honing machine shall be provided with not less than 10 feet of three-conductor flexible electrical cord, in accordance with NFPA-79 and NFPA-70, type SO. The cord shall have a voltage and current carrying capacity of not less than the voltage and current rating of the motor. The cord shall be provided with a three-prong grounding plug conforming to W-C-596/13.

3.6.3.4 High potential. The entire electrical system of the honing machine shall be capable of withstanding, without damage, malfunction, breakdown, arcing, surface or air discharge, the application of at least 1000 volts plus two times the operating voltage of the honing machine, 60 Hertz, sinusoidal wave electrical energy at standard temperature and pressure. The electrical energy shall be applied instantaneously at full voltage and shall be maintained for at least 60 continuous seconds. Each circuit shall be isolated from other circuits and all circuits shall be isolated from electrical ground. Solid state control components may be damaged as a result of high voltage and is not cause for rejection. Damage to solid state control components may be avoided by wiring around or by-passing them, terminal to terminal. No arcing or discharge from solid state control components is acceptable. To avoid excessive stressing of the insulation, repeated application of the high potential voltage is not allowed.

3.7 Dimensions and weight.

3.7.1 Dimensions. Unless otherwise specified (see 6.2), the overall dimensions of the honing machine shall not exceed 36 inches (914.4 mm) in length by 32 inches (812.8 mm) in width by 26 inches (660.4 mm) in height.

3.7.2 Weight. Unless otherwise specified (see 6.2), the weight of the honing machine, not including spare parts or accessories, shall not exceed 300 pounds (135.9 kilograms).

3.8 Honing heads or mandrels. Unless otherwise specified (see 6.2), hoing heads or mandrels supplied with each honingmachine shall include coarse and fine stones for honing through holes of any diameter within a range of 0.480 inch to 3.000 inches (12.192 to 76.2 mm), inclusive, and coarse and fine stones for honing blind holes or step cylinders of any diameter within a range of 1.125 inches to 2.062 inches (28.575 to 52.375 mm) inclusive.

3.9 Accessories. Unless otherwise specified (see 6.2), the following accessories shall be furnished with each honing machine:

a. One each, dressing fixture with diamond nib, or one complete set of truing sleeves.
b. one each stone dressing stick.
c. Five gallons, honing fluid.
d. One each, pin vise.
e. One each, carrying case, to store and transport all mandrels, accessories, spare parts and tools fabricated from either sheet steel or high density polyethylene.

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3.10 Spare parts. Unless otherwise specified (see 6.2), the following spare parts shall be furnished with each honing machine. Spare stones shall cover the size range specified in 3.1.

a. One each, spare drive belt.
b. One set, coarse stones for through holes.
c. Two sets, fine stones for through holes.
d. One set, coarse stones for blind holes or step cylinders
e. Two sets, fine stones for blind holes or step cylinders.

3.11 Tools. All tools, other than common hand tools, that are necessary for operation and maintenance of the honing machine, shall be provided.

3.12 Information plates. Nameplates, instruction plates, and lubrication plates shall be corrosion resistant metallic plates, printed in American English Language, and shall be permanently fastened to the honing machine in a location(s) visually convenient to the operator. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The plates shall be secured to the honing machine with permanent fasteners such as screws, bolts, or rivets at each of the four corners of the plain. The information plate(s) shall include warnings, cautions, and machine operator instructions to be followed when operating or maintaining the honing machine. The following information shall be included on the nameplate:

Nomenclature
National stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number

The following information shall be included on the lubrication plate:

Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal specification number of Lubricant

3.13 Workmanship Standards of workmanship shall assure that the honing machine shall have the stability, strength, safety, and efficient operating characteristics found in commercial honing machine and as specified in Section 3 herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All honing machines shall meet all requirements of sections 3 and 5. The inspection set forth herein shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements herein shall not relieve the contractor of the responsibility of assuring that all honing machines submitted to the government for acceptance cc

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with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When specified (see 6.2), first article inspection shall be applied to the preproduction model or initial production honing machine (see 3.1). The first article shall consist of the examination in 4.4, all tests under 4.5, and the packaging inspection and test under 4.6. Failure of the first article honing machine to pass all examinations and tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production honing machines offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of "a." through "c." below. Failure of any item to pass all examinations and tests shall be cause for rejection of the honing machine.

a. Product examination (see 4.4).
b. Tests (see 4.5).
c. Packaging inspection (see 4.6).

4.3 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or Test	Lot or Batch Size	Sample Size
Product examination (see 4.4)	1 - 150	13
Accuracy test (see 4.5.1)	1 - 150	13
Paint test (see 4.5.2)	1 - 150	13
High potential test (see 4.5.3)	1 - 150	5
Packaging test (see 4.6)	1 - 150	3
Rough handling test (see 4.6.1)	1/	1/

The lots shall not exceeed the maximum sizes indicated. If lot size is less than or equal to the sample size, 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if me or more defects are found.

1/ When first article test is specified, the packaging validation tests shall be performed on the first article rely. If no first article test is required, the rough handling test shall be performed on one unit pack selected randomly from the first lot of the total quantity of the honing machines supplied.

4.4 Product examination. Visually, dimensionally, and manually examine each honing machine to determine conformance with the requirements of 3.2 through 3.13 visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified herein. Manual examinations shall include the operation of movable parts by hard to assure proper functioning. The examination provisions may be

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applied at the earliest practical point in manufacturing at which it is feasible to inspect for acceptance without risk of change in the characteristics by subsequent operation. Failure of the contractor to provide objective evidence that the honing machine and its components have passed the examinations prescribed by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the honing machine shall be cause for rejection (see 3.2.3).

4.5 Tests.

4.5.1 Accuracy. The honing accuracy testshall be conducted on a high grade gray iron casting not less than 4 inches (101.6 mm) long with a hole drilled and reamed or bored axially to 1.000 inch (25.4 mm) diameter. The honing machine shall be used to finish the hole to 1.005 inches (25.527 mm), including both rough and finish honing operations. The finish honing operation shall involve the removalof 0.0005 to 0.0007 inch (.0127 mm to .01778 mm) of stock from the diameter. The accuracy of the entire length of the finished hole shall be within 0.0002 inch (.00508 mm) for roundness and taper. Final surface finish shall be 16 microinches (0.40 micrometers) maximum aa. Failure to meet required accuracies shall be cause for rejection (see 3.5.1).

4.5.2 Paint. A test sample, made of the same metal as the production honing machine, shall be tested for paint adhesion, hardness, and thickness. The sample shall receive the same surface treatment, the same thickness of pint, and be painted at the same time with the same paint as the production honing machine. The tests are described in a, b, and c following. The criteria for rejection are given in d. following:

a. Two intersecting straight lines, approximately two inches long, shall be cut through the paint to the base metal. A strip of fiberglass tape shall be pressed over the intersection and pulled from the surface.
b. The sharpened leads of an H and an HB wooden lead pencil shall be pressed vertically onto the painted surface until the leads crumble or break. The surface shall be cleaned and examined.
c. The thickness of the unpainted test sample shall be measuredand recorded to an accuracy of 0.0002 inch (.00508 mm). The measured value shall be used in the calculation of final paint thickness.
d. A paint thickness of less than 2.0 mils (0.0020) (.0508 mm), or an indentation caused by the HB lead, or no indentation left by the H lead, or evidence of any loss due to removal of the fiberglass tape, shall be cause for rejection (see 3.4.3).

4.5.3 High potential. The electrical circuits shall subjected to a test voltage of 1000 volts plus two times the rated voltage of the honing machine at 60 Hertz, sinusoidal wave electrical energy at standard temperature and pressure. This test voltage shall be applied between mutually isulated circuits, between insulated circuits and ground, and between each conductor and the chassis ground. The full test voltage shall be applied instantaneously and maintained for no less than 60 continuous seconds to each circuit in turn. During the 60 second intervals at the test voltage level, the circuitry shall be observed for flashover (surface discharge), sparkover (air discharge) and breakdown (puncture discharge). Any evidence of flashover, sparkover, breakdown or other damage to the honing machine due to heat or other effects of the applied voltage, shall be cause for rejection. Damage to solid state

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control components can be avoided by wiring around or by-passing the solid state control components, terminal to terminal. Arcing or discharge from solid state control components is unacceptable (see 3.6.3.4).

4.6 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

4.6.1 Rough handling. After the quality assurance inspection and tests have been successfully completed, one honing machine, packaged in accordance with the packaging requirements specified herein, shall be selected at random and subjected to the rough handling test (see 4.6.2). The apparatus used shall be as described in FED-STD-101. A dummy load shall not be used. All tests are to be conducted under ambient conditions. Environmental temperature and humidity tests are not required. After the test has been completed, the container shall be inspected. Defects in accordance with those described below shall be cause for rejection. If there is evidence of possible damage to the honing machine during the rough handlingtest, then the honing machine shall be unpacked and subjected to the productexamination (see 4.4) and accuracy test (see 4.5.1) to reaffirm that these requirements are still met, and then the honing shall be repackaged before submission for government acceptance. After the rough handling test has been completed, any damage to the packaging shall be replaced prior to acceptance of the lot.

4.6.2 Mechanical handling. The mechanical handling test shall be conducted using the lifting and transportation by forklift truck procedure of Method 5011.1 of FED-STD-101. Criteria for passing or failing the mechanical handling test shall be as follows:

"When the container or package is subjected to the mechanical handling test, the contents (except a dumy load) shall show no functional or physical damage, and the container and packing shall show no functional damage. Damage to the exterior shipping container which is the result of improper interior packaging, blocking, or bracing shall be cause for rejection. Structural failure of the exterior shipping container which would result in spilling of the contents or failure of the container in subsequent handling is cause for rejection. There shall be no evidence of a substantial amount of shifting of the contents within the exterior shipping container that would create conditions likely to cause damage during shipment, storage, and reshipment of the container. Minor container damage such as chipping of wood members, negligible dents, paint chipping, is not cause for rejection."

5 PACKAGING

5.1 First article. When specified (see 6.2), the supplier shall furnish one complete pack for first article inspection. Whenever first article inspection is specified or performed, unit pack examination and tests for packaging may also be performed (see 5.10 and 4.2.1). The first article may be either a preproduction pack or an initial production pack which conforms to the requirements of this specification. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish honing machines in accordance with the requirements of this specification.

5.2 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassmbled parts shall be preserved, anchored, braced, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled

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honing machine. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.3 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the honing machine shall be marked by stenciling, on the part, or by use of tags. Tags shall be waterproofed and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

5.4 Materials. Materials used in preservation-packaging and packing of the honing machine shall be as specified herein. Materials shall be new and free from all defects and imperfections which may affect the serviceability of the completed package or pack.

5.5 Processing conditions. Preservation-packaging shall be accomplished within buildings which have facilities to limit the accumulation of dust and moisture on the equipment being processed. Machinery brought into processing areas from outside shall be allowed sufficient time to reach ambient temperature before processing. Cleaning, drying, and application of preservatives shall progress in an uninterrupted series of operations. When periods of interruption are necessary, temporary protection shall be provided to partially processed items, as required, to insure against contamination.

5.6 General requirements. Component parts of the honing machine, accessories, and maintenance tools shall be cleaned, dried, preserved, packaged, and packed as specified herein. Parts and assemblies which are not covered by this specification shall be preserved and packaged utilizing the materials and methods specified for similar parts and assemblies described herein. Cleaning and drying procedures, and preservatives and their application, specified herein, shall conform to the criteria and guidelines of MIL-P-116. The fundamental principles and approved methods and techniques of packaging and packing are prescribed in MIL-P-116.

5.7 Preservation. Preservation shall be Level A or C as specified in the contract (see 6.2).

5.7.1 Level A.

5.7.1.1 Preservation-packaging. Unless otherwise specified, the basic honing machine shall be preserved and packaged as stated herein and in accordance with method I criterion of MIL-P-116. Electronic and electrical equipment, hand tools and accessories shall be preserved and packaged as specified herein. When the supplier determines that a preservative is required to protect a specific surface or item, but none has been specified herein, the preservative utilized shall be selected from MIL-P-116.

5.7.1.2 Cleaning and drying. Thorough cleaning and drying of all bare ferrous, unprotected, unfininshed, unpainted metal surfaces shall be accomplished before the application of a prservative. Care must be taken to limit the handling of cleaned parts prior to the assembly of new honing machines. Where practicable, processors and assemblers shall wear rubber or plastic gloves while handling cleaned parts. Unless otherwise specified herein, honing machines which have been fuctionally tested prior to shipment shall have all operational systems thoroughly drained before the application of a preservative. It may be necessary to connect the honing machine to a power source and operate at low speeds in order to drain all cavities. Extreme care shall be taken to immediately disconnect power when the honing machine is being drained so as not to damage pumps, motor or other components.

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5.7.1.3 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly of new honing machines shall be cleaned in accordance with process C-5 and dried in accordance with the applicable procedure of MIL-P-116. In the interim, to preclude contamination from forming on cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil conforming to P-9. To prevent dust from accumulating on cleaned parts, cover with a sheet of greaseproof material, such as MIL-B-121. In the event the supplier deems it necessary to clean a particular mechanism after draining the operational lubricant or a preservative. Internal mechanisms shall be cycled to assure all areas are protected with a coating of the lubricant before either sealing or draining. Any unprotected areas will have a preservative coating applied.

5.7.1.4 External surfaces. All accessories and attachments such as chucks, faceplates, steady rests, toolholders, abrasive wheels, etc. shall be removed from the assembly prior to the external cleaning operation. All external, critical ferrous metal surfaces, fit or function, of the honing machine shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. Noncritical ferrous metal surfaces shall be solvent cleaned and wiped or drain dried. Nonmetallic composites and painted metal surfaces shall be cleaned by wiping with a clean, dry cloth. Care must be taken to prevent cleaning solvents from coming in contact with nonmetallic composites. When cleaned surfaces of the honing machine show signs of rusting before application of the final preservative, the cleaning process shall be confined to those small areas of the assembly. Alternatively, the solvent used for the final wiping operation shall contain a small amount (approximately 5 percent by volume) preservative oil conforming to P-10, type 1, grade 30. It is extremely important that elapsed time between these operations be held to an absolute minimum.

5.7.1.5 Disassembled components. Disassembled components detached from the basic honing machine shall be cleaned and dried as specified herein for like items.

5.7.1.6 Maintenance and machine tools and tool accessories. All machine, maintenance, metal or woodworking, hand tools and any acessories thereof shall be preserved and packaged in accordance with level A requirement of PPP-P-40.

5.7.1.7 Electrical equipment. Electrical equipment, whether attached or detached, shall be cleaned, dried and preserved in accordance with Level A requirements of MIL-E-17555 or MIL-E-16298, whichever applies to the individual electrical item.

5.7.1.7.1 Electrical motors. Electrical motors, whether attached or detached, shall be preserved and packaged in accordance with Level A requirements of MIL-E-16298, and applicable subparagraph for assembled machines.

5.7.1.7.2 Electrical control devices. Control units, whether attached or detached, shall be preserved and packaged in accordance with Level Level A requirements of MIL-E-17555.

5.7.1.7.3 Electrical wiring system. Exposed ends of wires, plug openings, sockets, connector plugs, terminals and openings into switches and junction boxes shall be sealed with tape or cap-plugs conforming to PPP-T-60, type IV or MIL-C-5501, respectively. Electric cable shall be coiled to a minimum safe diameter and secured with cotton twine or tape to prevent uncoiling.

5.7.1.8 Hoses and fittings. Hoses and fittings which are not attached to the basic honing machine shall be cleaned, dried and preserved in accordance with Level A requirements of MIL-H-775.

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5.7.1.9 Lubricating systems. The honing machine, with all lubricants, greases and other lubricating compounds either installed or included shall have all vent openings in the lubrication systems temporarily capped, plugged or taped closed to prevent loss of lubricants during shipment and storage (see 3.2.2). Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps, plugs or tape intended to be vented to the atmosphere during normal operation. These warnings shall also be included in the instruction manual. In the event a lubrication reservoir cannot be covered and sealed, it shall be appropriately preserved and the lubricant shall be provided in a sealed container and shipped with the honing machine. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE MACHINE".

5.7.1.10 Grease fittings. Charge all grease fittings with preservative compound conforming to P-11.

5.7.1.11 Nonlubricated interior surfaces. Surfaces of internal gears which do not run in lubricant shall be thoroughly coated with P-7 preservative.

5.7.1.12 Gear cases. Fill the gear cases with supplier's normal lubricating oil. Shift gears into all possile positions while operating the honing machine. Make sure all surfaces are coated with lubricating oil and close and seal all openings to prevent any leakage during shipment and storage.

5.7.1.13 Sliding or friction surfaces. Surfaces of ways, gibs, driving gears, traversing and adjusting screws, and other critical, exterior operating machined surfaces shall be coated with P-2 preservative compound. Parts of the honing machine which are adjustable shall be locked into place by means provided and blocked to assure movement shall be kept to a minimum.

5.7.1.14 Non-critical surfaces. Exterior, non-critical, ferrous metal surfaces of the basic honing machine, which have not been painted or plated, shall be coated with
type P-2 preservative and covered. Such places as legs, bolts, etc. from which the preservative need not be removed may be preserved with P-1 or P-19.

5.7.1.15 Bearings and journals. Grease lubricated bearings shall be preserved with type P-11 preservative. Oil lubricated journals shall be filled with normal lubricating oils or with P-10, type I, grade optional.

5.7.1.16 Drive belts and pulleys. Drive belts shall either be removed from the honing machine or released from tension. Belts removed from the machine shall be packaged in accordance with method IC-1 or IC-3 criterion of MIL-P-116. Ferrous metal pulleys shall be coated with rust-inhibiting primer conforming to TT-P-664.

5.7.1.17 Indicators and meters. All indicators and meters shall be covered with a minimum of 1 inch thick uncompressed cushioning materials conforming to A-A-1898, grade II, class B. Secure cushioning material in place with tape conforming to PPP-T-60, type IV. Other type cushioning materials and tapes may be used provided the materials are water-resistant, non-corrosive, and will provide equivalent protection of items specified above.

5.7.1.18 Closure of openings. Small openings (except vents and louvers installed for ventilation purposes) which will admit dust or water shall be sealed with tape Conforming to PPP-T-60, type IV. Large openings shall be covered with waterproof

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barrier material conforming to class E-1 or L-4 of PPP-B-1055, or, as a minimum plastic material conforming to L-P-378, type and class optional. Secure materials in place with tape specified above. When very large openings are covered, or when location of openings render coverings vulnerable to puncture, coverings shall be protected by applying wood, plywood or sheetmetal over the materials. Open ends of all piping and fittings shall be plugged to prevent the entrance of foreign materials, using pipe fittings, plastic cap-plugs conforming to MIL-C-5501 or tape conforming to above.

5.7.1.19 Technical data. Technical data furnished with the honing machine shall be packaged in accordance with method IC-1 criterion of MIL-P-116. The package shall be marked "TECHNICAL DATA" in bold black leters. The package shall be protected against puncture and abrasion. The data shall be either attached to the honing machine with tape conforming to PPP-T-60, type IV, placed in a stowage compartment of the honing machine, or positioned in a consolidated box containing accessories or disassembled components.

5.7.1.20 Consolidated packaging. Packaged accessories, maintenance tools, small disassembled components, etc. shall be consolidated and placed in drawers and stowage compartments of the machine. When it is not feasible to stow parts as specified above, place parts in a box(es) conforming to PPP-B-636, class WR or WWVR; PPP-B-640, class 2; PP-B-601, domestic type; or PPP-B-621, class 1, in accordance with weight limitations specified.

5.7.1.21 Cushioning. Cushioning materials and methods of application shall conform to the criteria and guidelines of MIL-P-116 and MIL-STD-1186. In the event there are conflicting requirements, MIL-STD-1186 will take precedence.

5.7.1.22 Protection of preservatives. All exposed presered surfaces of the honing machine which is subject to being disturbed during the packing operation shall be loosely covered with barrier material conforming to MIL-B-121, type 1, grade A. Not less than two layers of similar-type barrier material shall be used to separate wood or metal blocking and finished metal surfaces of the honing machine. When considered necessary, secure barrier in place with tape conforming to PPP-T-60, type IV.

5.7.1.23 Movable parts. Operating heads, sliding tables, rams, counter-balances, and other movable parts shall be secured by mechanical devices incorporated in the honing machine. The devices alone do not always provide adequate support to prevent movement in the handling and transporting of the item and therefore must be augmented with appropriate wood blocking, metal tie rods used in conjuction with wood blocking, metal strapping, or other devices which will immobilize the part. Handwheels, cranks, levers, and similar controls shall be secured in position with common annealed wire or steel strapping.

5.7.1.24 Positioning assemblies. Assemblies having vertical movement shall be blocked and braced in a lowered position; those capable of moving traversely shall be centered for blocking and bracing.

5.7.2 Level C. Level C packaging/preservation shall be in accordance with the requirements and guidelines as specified in MIL-STD-1190.

5.8 Packing. Packing shall be level A or C as specified in the contract (see 6.2).

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5.8.1 Level A.

5.8.1.1 Machinery weighing less than 2,500 pounds. Unless otherwise specified, machinery including accessories, weighing less than 2500 pounds, shall be packed in cleated plywood boxes conforming to PPP-B-601, overseas type; nailed wood boxes conforming to PPP-B-621, class 2, style 2, 3, 4 or 7; or skidded boxes conforming to MIL-B-26195, type II, style A, class optional, with plywood superstructure. All shipping containers shall be provided with forklift capabilities. Selection of containers shall be governed by the size and weight limitations of the applicable container specification. Machines designed with mounting holes in areas that can withstand rough handling without breakage shall be bolted to an auxiliary base before being packed in a box conforming to PPP-B-601 or PPP-B-621. Contents of boxes shall be arranged, anchored, blocked, braced and cushioned in accordance with MIL-P-116 and **MIL-STD-1186. In the event there are conflicting requirements, MIL-STD-1186 will take precedence. Box closure and strapping requirements shall be in accordance with the appendix to the applicable box specification.**

5.8.1.2 Consolidated containers. Consolidated containers as specified shall be placed inside the shipping container housing the honing machine. The boxes shall be restrained by the use of metal strapping of sufficient strength or tie rod assemblies.

5.8.1.3 Operational oils. Operational oils furnished with the machine, but not placed in the honing machine stowage receptacles, shall be stowed in metal drum(s) conforming to PPP-D-705, type I or II or PPP-D-729, type I or II. Drums shall be positioned upright inside the shipping container housing the honing machine. Sufficient blocking shall be applied to immobilize the drum(s) inside the shipping container. Alternatively, when the drum(s) tend to increase the cubage of the shipping container, the drum(s) may be shipped as individual containers.

5.8.2 Level C. Level C packing shall be in accordance with the requirements and guidelines as specified in MIL-STD-1190.

5.9 Marking. All marking shall be in accordance with MIL-STD-129. Bar code markings are required and shall be in accordance with MIL-STD-129. Center of balance markings and arrow markings, indicating THIS SIDE UP, shall be applied to each shipping container for all levels for the protection of the honing machine and personnel.

5.10 Rough handling. The completed unit pack, shipping container(s) and all contents shall be capable of withstanding the shipping and handling conditions expected from fork lift handling and truck transport. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test specified herein.

5.11 Workmanship. Workmanship shall be such that when the proper procedures are fond, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration consistent with the shipment hazards known or anticipate, and suitable for the term of storage expected.

5.12 Material safety data sheet (MSDS). MSDSs shall be included for any and all products deemed to be an irritant or to have a possible impact on the environment or public and personnel safety. These items shall include, but not be limited to such things as liquids, powders, sprays, cleaning solutions, paints, etc.

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6 NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The honing machine covered by this specification is intended for use in honing bushings, bearings, and hydraulic brake cylinders, including step cylinders, by either wet or dry methods, within a diameter range of 0.480 to 3.000 inches, inclusive, for through holes; and a diameter range of 1.125 through 2.062 inches, inclusive, for blind holes and step cylinders.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).
c. First article, when required (see 3.1, 4.2.1 and 5.1).
d. Electrical equipment, if different (see 3.6.3).
e. Motor requirements, if different (see 3.6.3.1)
f. Length of cord, if different (see 3.6.3.3).
g. Dimensions, if different (see 3.7.1).
h. Weight, if different (see 3.7.2).
i. Size range of honing heads or mandrels, if different (see 3.8).
j. Accessories required, if different (see 3.9).
k. Spare parts, if different (see 3.10).
l. Quality conformance inspection, if different (see 4.2.2).
m. Preservation and packing when required (see 5.7 and 5.8).

6.3 First article. When a first article inspection is required the honing machine(s) should be a first article sample, a first production item, a sample selected from the first run of production honing machines or it may be a standard production honing machine from the contractor's current inventory as indicated. The first article should consist of a specific number of honing machines. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering products which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identif on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.5 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, and price. Repai part identification should also include the name of the original equipment/part manufacturers, Contractor and Government Entity (CAGE), nomenclature, and part number immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

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6.6 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and controls of noise levels, radiation, electromagnetic emissions, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.7 Subject term (key word) listing.

Bearing
Brake Cylinder
Bushing
Dry
Hone
Horizontal
Wet

6.8 Changes from the previos issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army -AL

Reviewer:
DLA - GS

User:
Army - ME

Preparing activity:
Army - AL

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INSTRUCTIONS

1. The preparing activity must complete blocks 1,2,3 and 8. In block 1, both the document number and revision letter should be given.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-14253F	2. DOCUMENT DATE (YYMMDD) 940610

3. DOCUMENT TITLE
HONING MACHINE, HORIZONTAL, WET TYPE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a.NAME (Last, First, Middle initial)	b.ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1)COMMERCIAL (2) DSN (if applicable)	7.DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander	b. TELEPHONE (Include Area Code) (1) Commercial (2) DSN
c. ADDRESS (Include Zip Code) US Army Research, Development and Engineering Center ATTN: SMCAR-EST-P Rock Island, IL 61299-7300	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Tel. (703)756-2340 DSN 289-2340

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